

Architecture Justification

Food Bank Surplus Redistribution Platform

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Architecture Selection

We chose **Layered Architecture** for the Food Bank Surplus Redistribution Platform. The architecture separates the system into Presentation, Business, and Data layers, providing clear separation of responsibilities. This follows the architectural style described in the Lab 3 handout.

Architectural Choice – Reason 1

The platform performs different functions such as managing food donations, matching surplus food with food-bank requests, and managing distribution. A layered structure separates these functions into appropriate components, making the system easier to understand and maintain.

Architectural Choice – Reason 2

The system requires interaction with different users such as donors, food banks/NGOs, volunteers, and administrators. Keeping the user interface separate from the business logic allows changes to the interface without directly affecting the core food redistribution functions.

Security Advantage

The layered architecture can restrict direct access to the database by routing requests through the business layer. This provides a place to perform validation and authorization before accessing user, donation, request, and distribution data.

Performance Benefit

Separating the business and data layers allows frequently used operations, such as retrieving available surplus food and processing requests, to be handled by dedicated business components before accessing the database. This can reduce unnecessary database operations and improve system response time.

Conclusion

Layered Architecture provides a structured design for the Food Bank Surplus Redistribution Platform by separating user interaction, business operations, and data storage. It supports maintainability, controlled data access, and efficient processing while keeping the components organized and clearly connected.